

FAKE CURRENCY DETECTION AND REPORTING SYSTEM

1. Title

FAKE CURRENCY DETECTION AND REPORTING SYSTEM

1. Abstract

Fake Currency Detection and Reporting System is a web-based application developed to assist users in identifying counterfeit currency notes and reporting suspicious cases to authorized personnel. The system provides a secure platform where users can upload images of currency notes for verification. The uploaded notes are analyzed and classified as Genuine or Counterfeit.

The application consists of three primary user roles: Admin, User, and Police. Users can register, log in, upload currency images, view detection results, and provide feedback. The Admin manages users, monitors uploaded currency details, and oversees the entire system. The Police module allows law enforcement personnel to view counterfeit currency reports along with the details of users who submitted them.

The system aims to reduce the circulation of fake currency by providing a centralized platform for currency verification and monitoring. It improves transparency, simplifies reporting, and supports authorities in tracking counterfeit currency activities.

1. System Architecture

Architecture Overview

Layer	Technology
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Django (Python)
Database	MySQL
Currency Verification	Image Upload & Detection Module
Reporting	Admin & Police Monitoring

System Flow

- Users register and log in to the system
- Users upload currency note images
- System analyzes the uploaded currency

- Detection result is generated (Genuine / Counterfeit)
 - Results are stored in the database
 - Admin monitors uploaded currency records
 - Police can view suspicious currency reports and user details
 - Users can provide feedback regarding the system
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1. Modules

The Fake Currency Detection and Reporting System consists of three major modules:

4.1 Admin Module

Purpose

To manage the entire system, monitor currency submissions, and oversee user activities.

Main Features

- Admin login
- View registered users
- Manage user accounts
- View uploaded currency details
- Monitor detection results
- View user feedback
- Remove unwanted records
- Monitor counterfeit currency reports

Functional Flow

- Admin logs in
- Views registered users

- Monitors uploaded currency notes
 - Reviews detection results
 - Checks suspicious currency submissions
 - Views feedback submitted by users
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4.2 User Module

Purpose

To allow users to verify currency notes and obtain detection results.

Main Features

- User registration
- User login
- Upload currency note images
- View uploaded note history
- Check currency detection results
- Provide feedback
- Manage profile details

Functional Flow

- User registers and logs in
- Uploads a currency note image
- System processes the image
- Result is displayed as Genuine or Counterfeit
- User views previous uploads
- User sends feedback

- User updates profile information
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4.3 Police Module

Purpose

To help law enforcement authorities monitor counterfeit currency cases and identify users who submitted suspicious notes.

Main Features

- Police login
- View uploaded currency details
- View counterfeit currency reports
- View user details associated with uploaded notes
- Monitor suspicious activities

Functional Flow

- Police logs in
 - Views currency records
 - Reviews counterfeit currency detection results
 - Accesses details of users who uploaded suspicious notes
 - Takes necessary action based on reports
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1. Technologies Used

Category	Technology
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Django (Python)
Database	MySQL
Authentication	Django Session Authentication

Category	Technology
Image Handling	Django File Upload
Development Tools	VS Code
Version Control	GitHub

1. Future Enhancements

- Integration with advanced AI/ML models for real-time currency detection
- Mobile application for Android and iOS
- Automatic notification to police for counterfeit detections
- OCR-based serial number extraction
- Currency denomination recognition
- Real-time analytics dashboard
- Email and SMS alert system
- Integration with government counterfeit currency databases

1. Conclusion

The Fake Currency Detection and Reporting System provides an effective solution for identifying and monitoring counterfeit currency. The system enables users to verify currency notes, allows administrators to supervise records, and assists police authorities in tracking suspicious activities. By automating currency verification and reporting processes, the application reduces manual effort, enhances transparency, and contributes to the prevention of fake currency circulation. The project demonstrates the practical implementation of Django and MySQL in developing a secure and scalable web-based detection and monitoring platform.